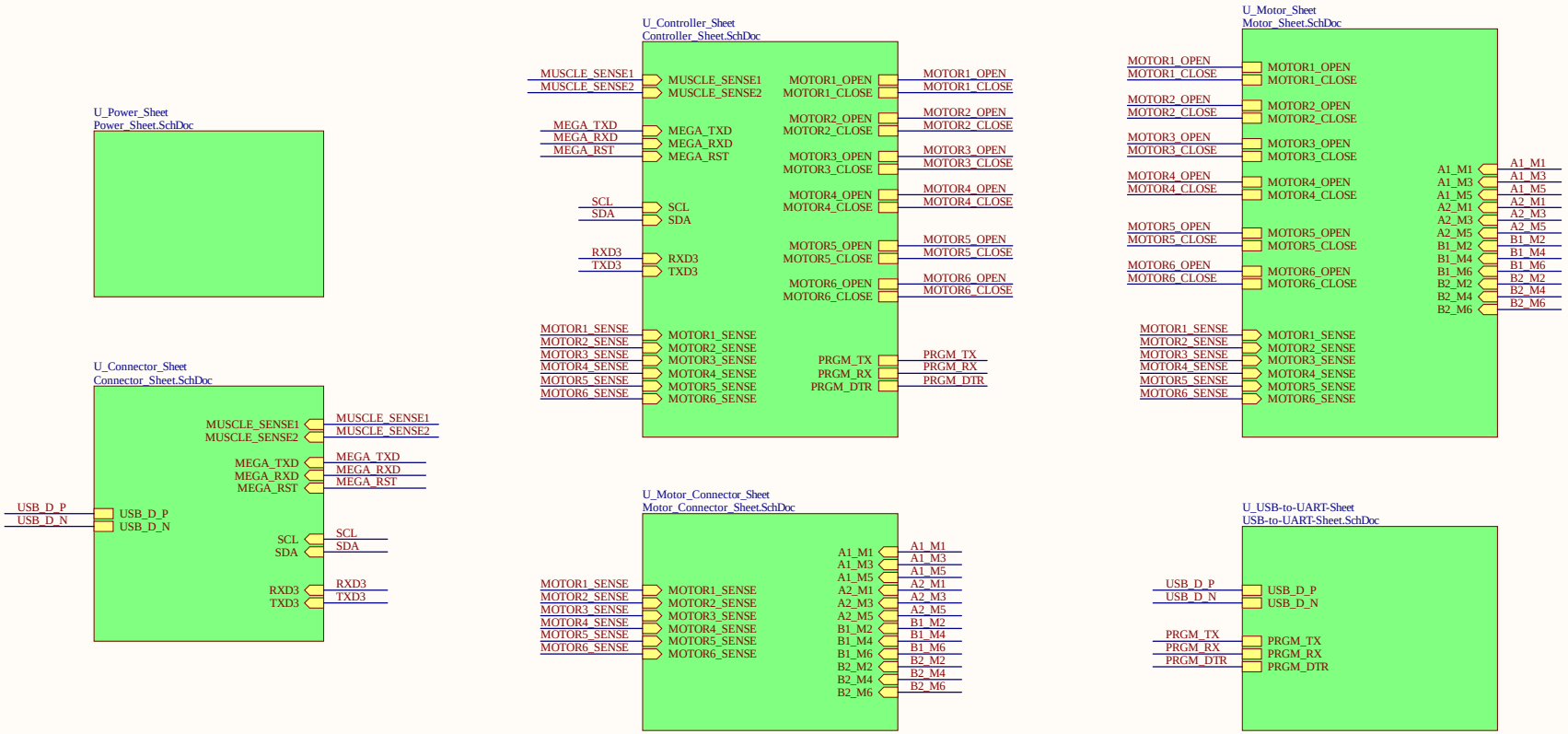
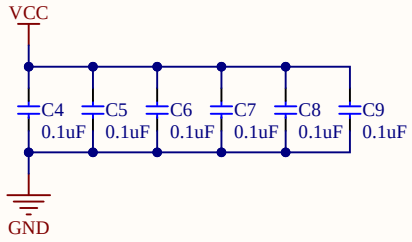
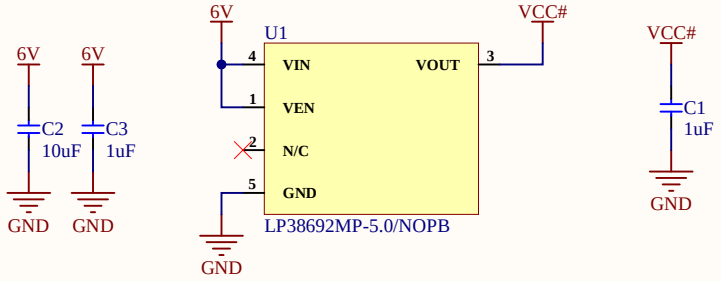

U_Power_Sheet
Power_Sheet.SchDoc
U_Connector_Sheet
Connector_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
MEGA_TXD
MEGA_RXD
MEGA_RST
SCL
SDA
RXD3
TXD3
USB_D_P
USB_D_N
U_Controller_Sheet
Controller_Sheet.SchDoc
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
PRGM_TX
PRGM_RX
PRGM_DTR
U_Motor_Sheet
Motor_Sheet.SchDoc
A1_M1
A1_M3
A1_M5
A2_M1
A2_M3
A2_M5
B1_M2
B1_M4
B1_M6
B2_M2
B2_M4
B2_M6
U_Motor_Connector_Sheet
Motor_Connector_Sheet.SchDoc
U_USB-to-UART-Sheet
USB-to-UART-Sheet.SchDoc

USB_D_N
USB_D_P

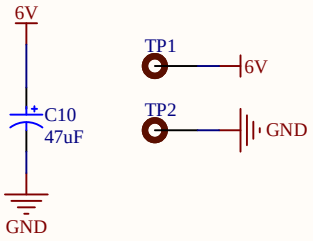
DECOUPLING CAPACITORS

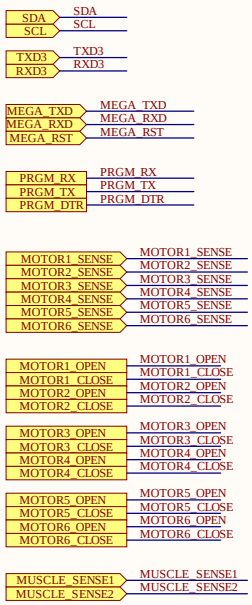


6V TO 5V CONVETER

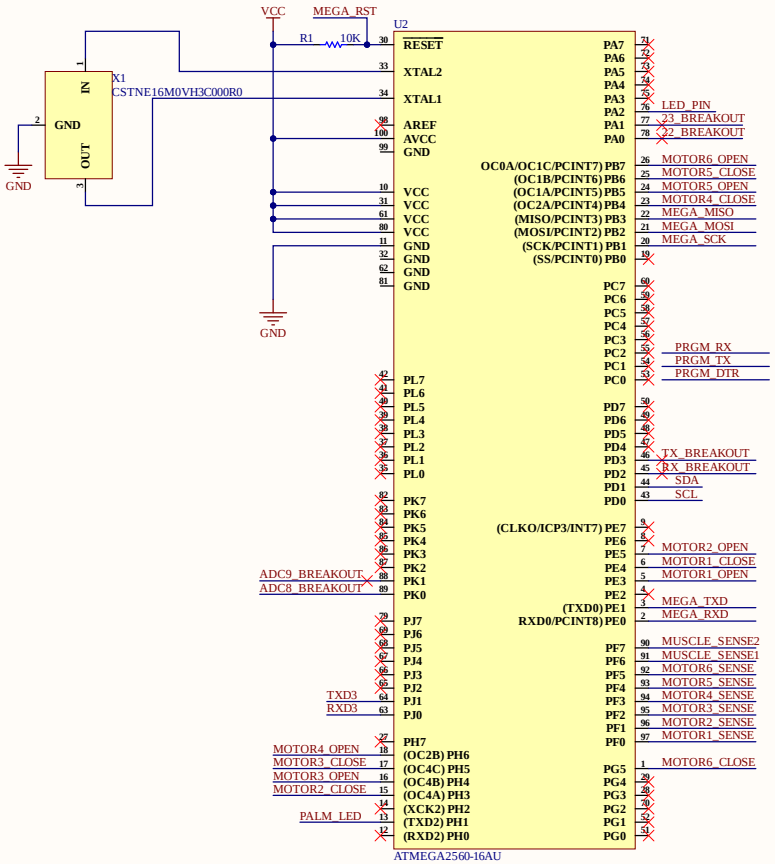


POWER INPUTS

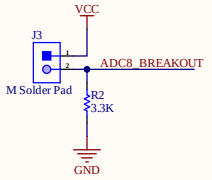




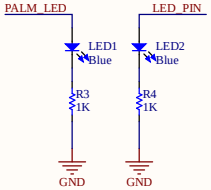
MAIN CONTROLLER



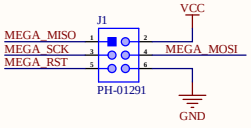
SENSOR CONNECTION

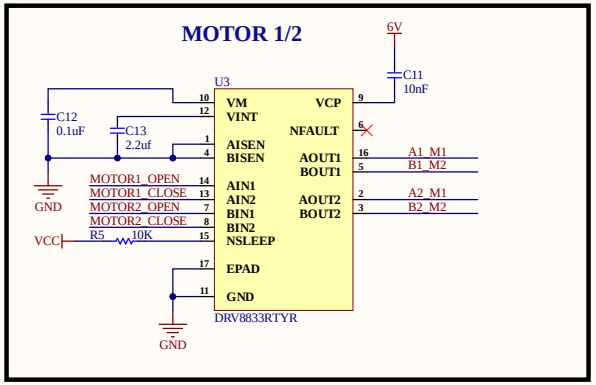


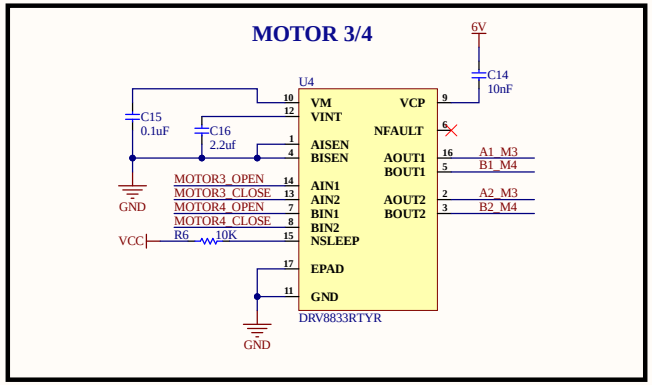
LED'S

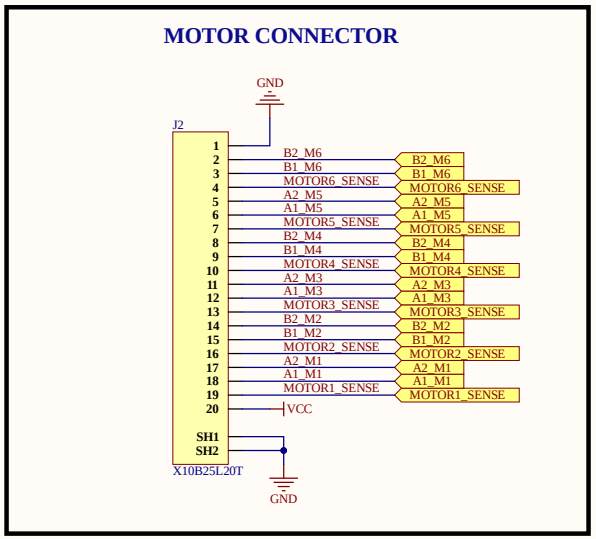


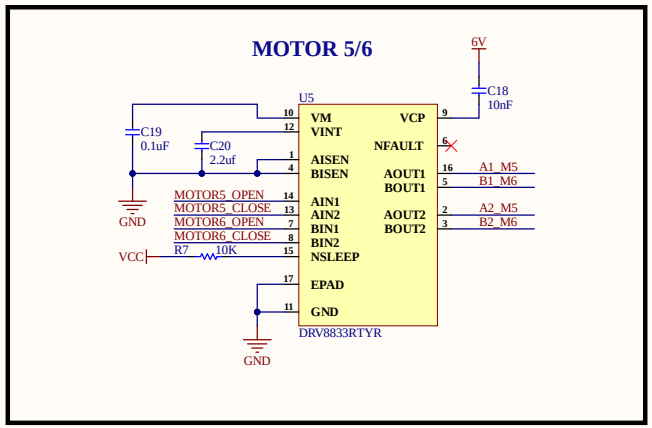
SPI

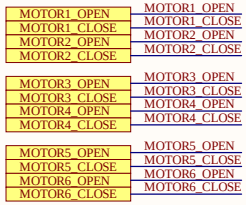


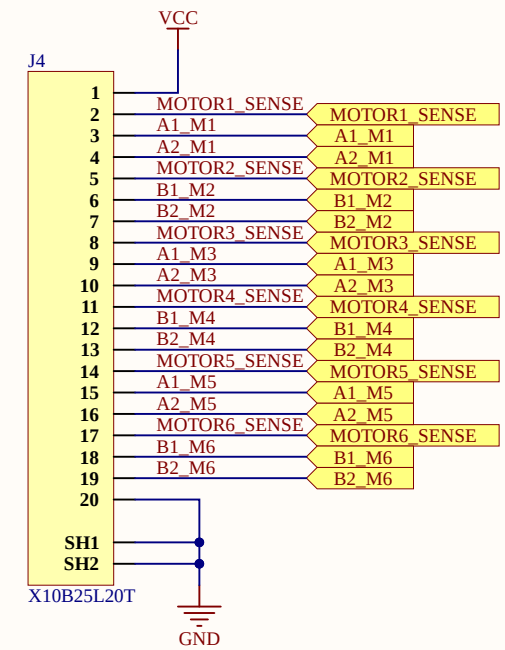

MOTOR 1/2
6V
C11
10nF
U3
C12
0.1uF
C13
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
A1_M1
B1_M2
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
AIN1
AIN2
BIN1
BIN2
AOUT2
BOUT2
A2_M1
B2_M2
GND
VCC
R5
10K
NSLEEP
EPAD
GND
DRV8833RTYR

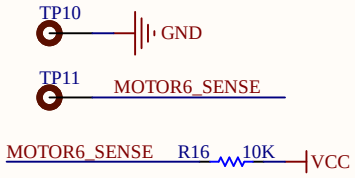

MOTOR 3/4
6V
C14
10nF
U4
C15
0.1uF
C16
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
A1_M3
B1_M4
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
AIN1
AIN2
BIN1
BIN2
AOUT2
BOUT2
A2_M3
B2_M4
GND
VCC
R6
10K
NSLEEP
EPAD
GND
DRV8833RTYR

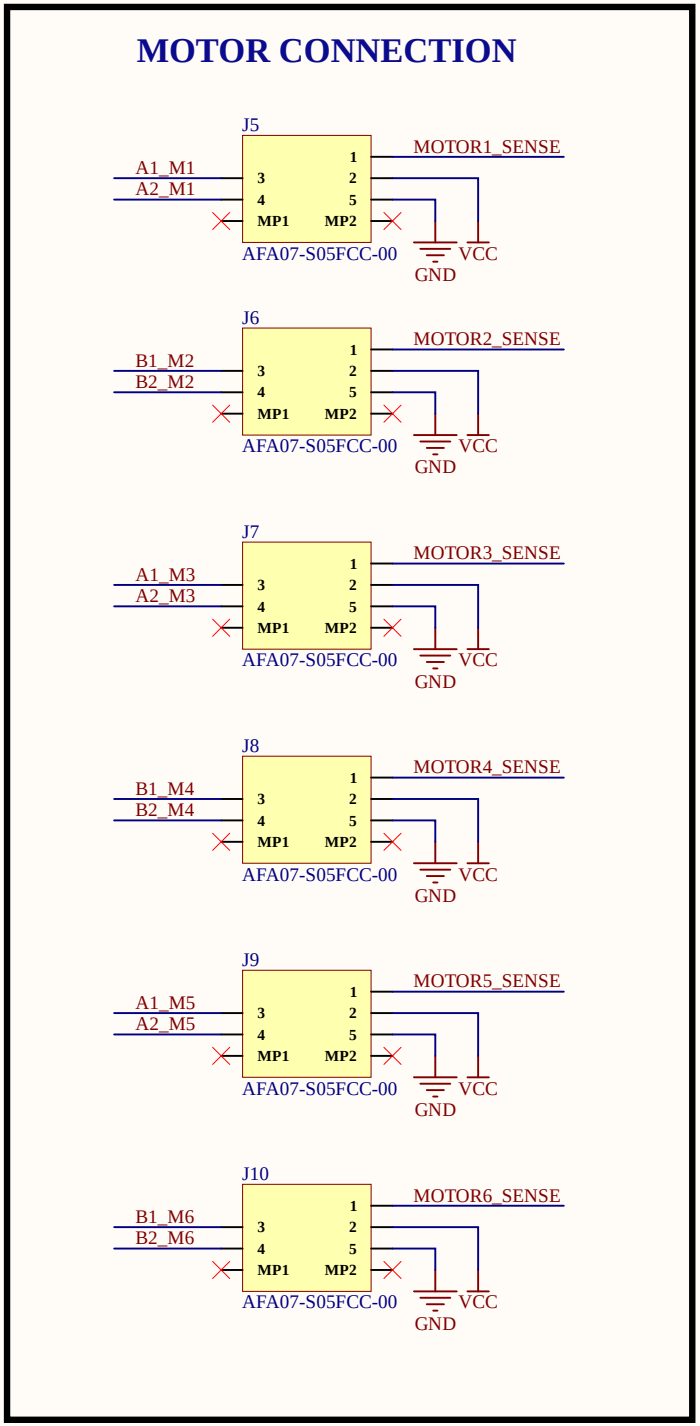

MOTOR CONNECTOR
GND
J2
B2_M6
B1_M6
MOTOR6_SENSE
A2_M5
A1_M5
MOTOR5_SENSE
B2_M4
B1_M4
MOTOR4_SENSE
A2_M3
A1_M3
MOTOR3_SENSE
B2_M2
B1_M2
MOTOR2_SENSE
A2_M1
A1_M1
MOTOR1_SENSE
VCC
SH1
SH2
X10B25L20T
GND


MOTOR 5/6
6V
C18
10nF
U5
C19
0.1uF
C20
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
A1_M5
B1_M6
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
AIN1
AIN2
BIN1
BIN2
AOUT2
BOUT2
A2_M5
B2_M6
GND
VCC
R7
10K
NSLEEP
EPAD
GND
DRV8833RTYR

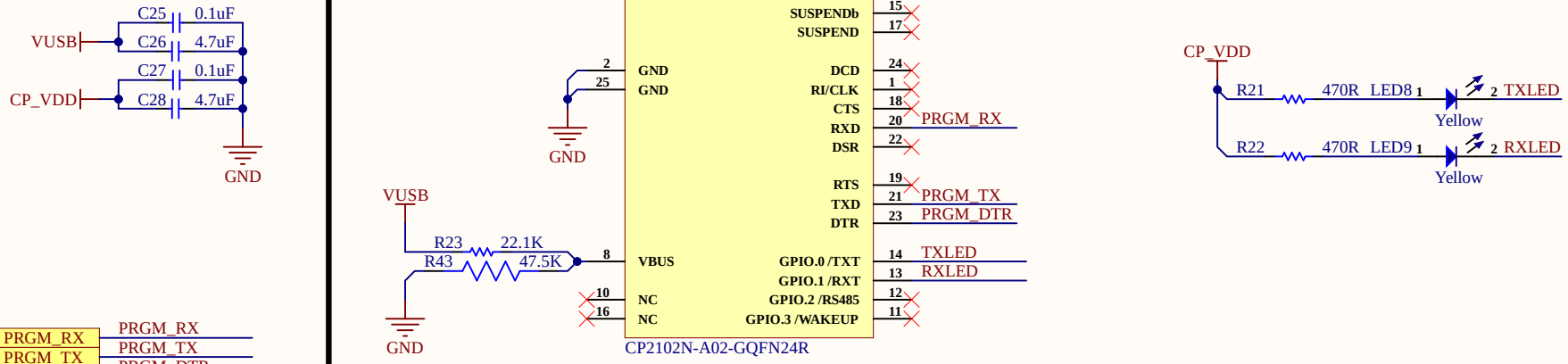

MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE


VCC
J4
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
SH1
SH2
X10B25L20T
MOTOR1_SENSE
A1_M1
A2_M1
MOTOR2_SENSE
B1_M2
B2_M2
MOTOR3_SENSE
A1_M3
A2_M3
MOTOR4_SENSE
B1_M4
B2_M4
MOTOR5_SENSE
A1_M5
A2_M5
MOTOR6_SENSE
B1_M6
B2_M6
GND

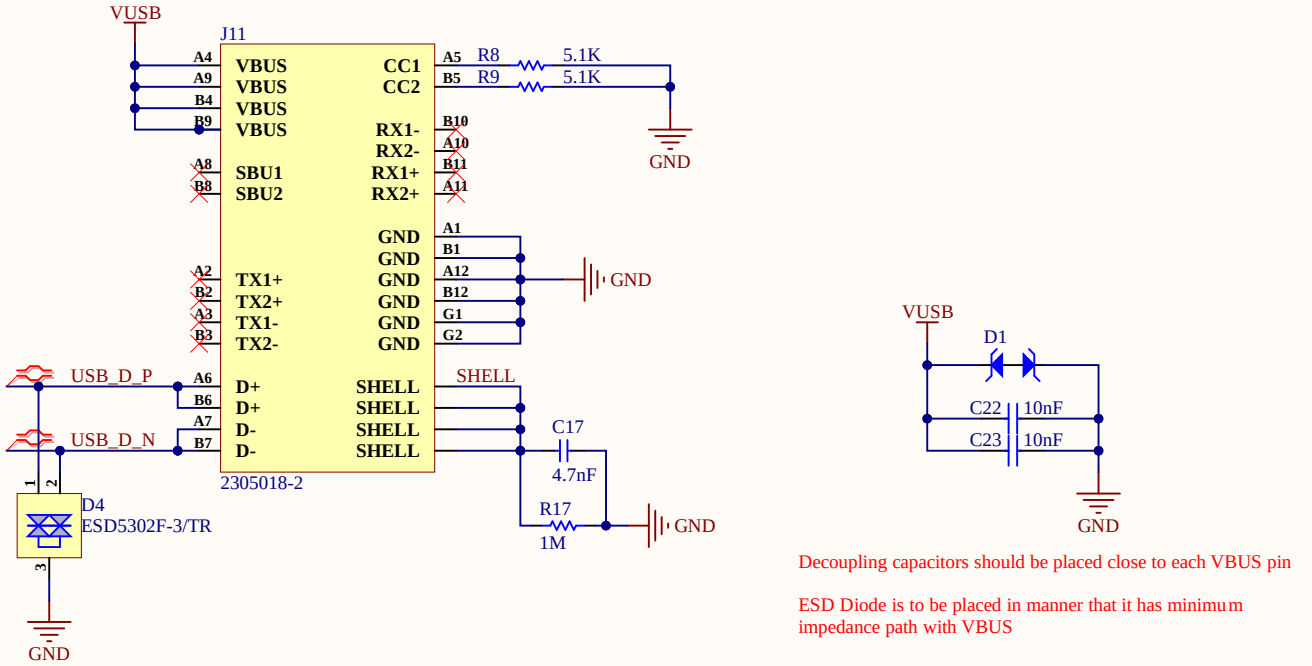

TP10
GND
TP11
MOTOR6_SENSE
MOTOR6_SENSE
R16
10K
VCC


MOTOR CONNECTION
J5
A1_M1
A2_M1
MOTOR1_SENSE
AFA07-S05FCC-00
VCC
GND
J6
B1_M2
B2_M2
MOTOR2_SENSE
J7
A1_M3
A2_M3
MOTOR3_SENSE
J8
B1_M4
B2_M4
MOTOR4_SENSE
J9
A1_M5
A2_M5
MOTOR5_SENSE
J10
B1_M6
B2_M6
MOTOR6_SENSE

USB-to-UART for Programming

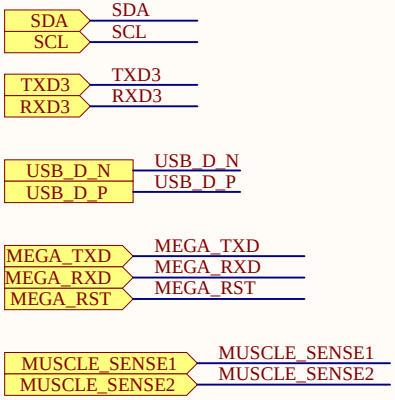


USB-C CONNECTOR

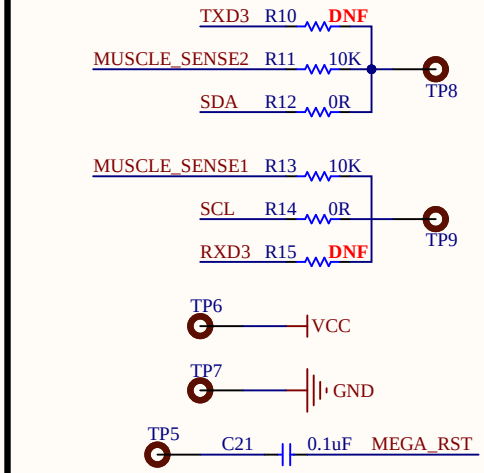


Decoupling capacitors should be placed close to each VBUS pin

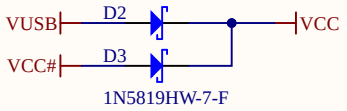
ESD Diode is to be placed in manner that it has minimum impedance path with VBUS



AUDIO CONNECTION



DIODE ORING



USB CONNECTION

